PROJECT REPORT

F1 RACE INTELLIGENCE

Formula 1 Race Performance Analysis
Using Tableau and Web Technologies

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Course: B.Tech Computer Science Engineering
Specialisation: Data Science
University: Medi-Caps University, Indore
Project Type: Data Analytics
Technologies: Tableau, HTML5, CSS3

GitHub Repository
https://github.com/XNXSHH/F1-Race-Analysis

1. ABSTRACT

The F1 Race Intelligence project is a data analytics and visualisation platform designed to explore Formula 1 racing statistics and performance trends. Formula 1 generates a large volume of historical racing data, including driver performances, race victories, team achievements and season-wise statistics. Analysing this information manually can be difficult, making data visualisation an effective approach to understanding racing trends.

This project uses Tableau to transform racing data into interactive visualisations and analytical dashboards. A dedicated website developed using HTML and CSS provides a structured and user-friendly interface through which users can access the dashboard and its accompanying data story. The project combines data analytics, visual storytelling and web development to present Formula 1 statistics in an organised and accessible format.

2. INTRODUCTION

Formula 1 is one of the most data-intensive motorsport competitions in the world. Every race produces valuable information that can be used to evaluate driver performance, compare teams and understand historical racing trends.

The F1 Race Intelligence project focuses on presenting this information through interactive dashboards and a dedicated website. The project consists of two major components:

- Data Visualization: Tableau is used to analyse Formula 1 statistics and create interactive dashboards and visual stories.
- Web Development: HTML and CSS are used to develop a website that integrates the Tableau visualisations and provides an organised browsing experience.

3. OBJECTIVES

- To analyze historical Formula 1 racing data.
- To understand driver performance and race-winning patterns.
- To examine season-wise race victories and team achievements.
- To convert raw statistical information into meaningful visualisations.
- To develop interactive dashboards using Tableau.
- To integrate Tableau dashboards into a functional website.
- To demonstrate the practical application of data analytics and web development.
- To communicate analytical findings through visual storytelling.

4. TECHNOLOGIES AND TOOLS USED

4.1 Tableau

Tableau is the primary data visualisation tool used in this project. It transforms structured datasets into interactive charts, dashboards and analytical stories. Users can explore different aspects of Formula 1 performance through visual representations rather than relying entirely on raw numerical data.

4.2 HTML5

HTML5 is used to create the structure of the website. It defines the different sections, including the navigation bar, homepage and introduction, dashboard section, racing story section and footer.

4.3 CSS3

CSS3 is responsible for the website's appearance and layout. It is used for the dark-themed visual design, typography, spacing, navigation styling, background images and gradients, hover effects and responsive layouts.

4.4 GitHub

GitHub is used for hosting and maintaining the project's source code. It provides access to the HTML implementation and allows the project to be shared publicly.

5. PROJECT METHODOLOGY

Stage 1: Data Understanding

The project begins with exploring Formula 1 racing information and identifying relevant performance indicators, such as race victories, driver achievements, team performance and season-wise statistics.

Stage 2: Data Analysis

The relevant information is organised and analysed to identify patterns and comparisons across different drivers, teams and racing seasons.

Stage 3: Data Visualization

Tableau is used to create charts and interactive visualisations. These visualisations help represent racing statistics in a clear and understandable format.

Stage 4: Dashboard Development

The individual visualisations are organised into a consolidated dashboard to provide a unified analytical experience.

Stage 5: Web Development

A dedicated website is developed using HTML5 and CSS3. It includes navigation, an introductory section, dashboard integration and a racing story section.

Stage 6: Integration and Deployment

The Tableau dashboard and story are embedded into the website using HTML iframe elements. The website source code is maintained in a GitHub repository for accessibility and sharing.

6. SYSTEM DESIGN AND IMPLEMENTATION

6.1 Data Analytics Module

The data analytics module is developed using Tableau. It contains interactive visualisations that explore different dimensions of Formula 1 racing performance, including:

- Driver performance analysis
- Race wins across different seasons
- Team achievements and comparisons
- Historical racing trends

The dashboard provides a consolidated view of the analysed information, while the Tableau story presents the visual analysis in a more structured narrative format.

6.2 Website Development Module

The website is titled F1. ANALYSIS and follows the concept of exploring Formula 1 beyond the finish line. Its major sections include:

A. Navigation Bar

The navigation bar provides three main navigation options: Home, Dashboard and Story. These links allow users to move between different sections of the website.

B. Hero Section

The homepage contains the heading “BEYOND THE FINISH LINE” with a motorsport background image. It introduces the project's purpose and provides a button that directs users to the analysis dashboard.

C. Introduction Section

This section introduces the concept of data-driven Formula 1 analysis and explains how visualisation can help understand motorsport performance.

D. Race Analysis Dashboard

This section embeds the Tableau dashboard directly into the website through an iframe. It allows users to explore the visualised racing statistics without navigating away from the project website.

E. Racing Story Section

This section integrates the Tableau story, providing a structured presentation of the analytical visualisations.

F. Footer

The footer displays the project title and project attribution.

7. WEBSITE DESIGN AND USER INTERFACE

The website follows a modern, motorsport-inspired design approach. Its visual elements include:

- Dark backgrounds for a professional appearance.
- Red highlights inspired by Formula 1 branding.
- High-contrast typography for readability.
- Background imagery related to motorsport.
- Section-based navigation for easy access.
- Responsive styling for smaller screen sizes.

The website uses CSS Flexbox for layout alignment and CSS media queries to adapt the navigation and content spacing to smaller screens. Smooth scrolling and hover transitions improve the overall browsing experience.

8. RESULTS AND OBSERVATIONS

The project combines Tableau-based data visualisation with front-end web development. The major outcomes include:

- Formula 1 racing statistics are represented through interactive visualisations.
- Driver and team performance can be explored through graphical comparisons.
- Season-wise race victories provide a historical perspective on racing achievements.
- The Tableau dashboard offers a consolidated analytical view.
- The Tableau story provides an additional visual storytelling experience.
- The website creates a centralised platform for accessing the project's analytical content.

The integration of Tableau with HTML and CSS demonstrates how data analytics outputs can be presented through a dedicated web interface.

9. CONCLUSION

The F1 Race Intelligence project demonstrates the practical application of data analytics and web development in the field of sports analysis. By using Tableau for interactive data visualisation and HTML5 and CSS3 for website development, the project presents Formula 1 racing statistics through an organised and accessible interface.

The project highlights the importance of transforming raw data into meaningful information and demonstrates how visual storytelling can improve the understanding of complex datasets. Overall, this project provides practical experience in data visualisation, analytical thinking, front-end development and the integration of external visualisation platforms into websites.

10. REFERENCES

1. GitHub Repository: https://github.com/XNXSHH/F1-Race-Analysis
2. Tableau Public: https://public.tableau.com/app/profile/anash.shah/vizzes